

Exp.No.6-File Modification Detection Using File Integrity Monitoring in Kali Linux

Aim:-

To detect unauthorized file modifications by generating and verifying file hashes using File Integrity Monitoring (FIM) techniques in Kali Linux.

Objective

- To understand the concept of File Integrity Monitoring
- To generate hash values of files
- To detect file modification by comparing hash values
- To analyze integrity violation

```
Session Actions Edit View Help
(unknown@kali)~$ mkdir -p ~/FIM_LAB && cd ~/FIM_LAB
(unknown@kali)~/FIM_LAB$ echo "This is the original file content" > testfile.txt
(unknown@kali)~/FIM_LAB$ cat testfile.txt
This is the original file content
(unknown@kali)~/FIM_LAB$ sha256sum testfile.txt > baseline_hash.txt
(unknown@kali)~/FIM_LAB$ cat baseline_hash.txt
4f1a14cd99a8e64e4d74e5f9f3a0b3e41a3aa8f91b56e3ea5d041319860145a2 testfile.txt
(unknown@kali)~/FIM_LAB$ echo "File has been modified " >> testfile.txt
(unknown@kali)~/FIM_LAB$ cat testfile.txt
This is the original file content
File has been modified
(unknown@kali)~/FIM_LAB$ sha256sum testfile.txt > new_hash.txt
(unknown@kali)~/FIM_LAB$ diff baseline_hash.txt new_hash.txt
1c1
< 4f1a14cd99a8e64e4d74e5f9f3a0b3e41a3aa8f91b56e3ea5d041319860145a2 testfile.txt
> 53ff0c07789b405934151a99258d946af903c4bec3bf26e12d80437341dbbb6 testfile.txt
(unknown@kali)~/FIM_LAB$
```